

DOMAIN KNOWLEDGE DOCUMENT

Social Media Advertisement Performance Dataset

1. About the Data

Dataset Kaggle: [Social Media Advertisement Performance](#)

This dataset represents Meta Ads Performance Data, covering campaigns, ads, user demographics, and ad interaction events. It is modelled after how Facebook/Instagram (Meta) ad platforms capture data.

The purpose of this dataset is to analyse advertising performance, optimize targeting, and measure ROI (Return on Investment) through KPIs such as:

- Impressions (how often ads are seen)
- Clicks (engagement with ads)
- Purchases (conversions)
- CPM, CPC, CTR, and ROAS (efficiency metrics)
- Audience insights (demographics, location, interests)

This dataset is ideal for building a Power BI Dashboard to evaluate campaign effectiveness, budget utilization, and user engagement patterns.

2. Use of Each Table

2.1. Table 1: ad_events

- Stores event-level logs (like impressions, clicks, purchases).
- This is the fact table in the model because all KPIs are derived from events.
- Used to analyze when and how users interact with ads.

2.2. Table 2: ads

- Contains details of each ad creative.
- Defines targeting criteria and which campaign an ad belongs to.
- Used for platform-level and creative-type-level analysis (e.g., Facebook Video Ads vs Instagram Image Ads).

2.3. Table 3: campaigns

- High-level campaign strategy and budget allocation.
- Provides timeframe and budget for ads.
- Used to calculate cost-based KPIs (CPC, CPM, ROAS).

2.4. Table 4: users

- Stores demographic and interest information of users who interact with ads.
- Used for audience segmentation (gender, age, country, location, interests).
- Helps analyze targeting efficiency (are ads reaching the right audience?).

3. Table and Field Details

Table 1: ad_events

Purpose: Captures every interaction (event) between a user and an ad.

Usage: This is the foundation for KPIs such as Impressions, Clicks, CTR, Conversion Rate, and ROAS.

Field	Description	Example	Use in Analysis
event_id	Unique identifier for each event	12	Used as primary key for the table
ad_id	Links to ads table	94	Join with ads → get ad_platform, ad_type
user_id	Links to users table	fa7e7	Join with users → get demographics
timestamp	Exact time of event	2025-07-21 03:10:13	Build date hierarchy (Day, Week, Month)
day_of_week	Derived field: day of the week	Monday	To compare weekday vs weekend performance
time_of_day	Derived field: segment of day	Night	See when users engage most
event_type	Type of event: Impression, Click, Share, Comment, Purchase	Impression	Funnel analysis (Impressions → Clicks → Purchases)

Table 2: ads

Purpose: Defines ad-level metadata.

Usage: Helps identify which platform + ad type combination works best, and whether targeting matches actual user engagement.

Field	Description	Example	Use in Analysis
ad_id	Unique ad identifier	51	Primary key; joins to ad_events
campaign_id	Campaign association	26	Join to campaigns table
ad_platform	Platform where ad runs (Facebook, Instagram, Messenger, Audience Network)	Instagram	Compare platform performance
ad_type	Creative format (Image, Video, Carousel, Story)	Video	Performance by creative type
target_gender	Gender targeted	Female	Check targeting efficiency
target_age_group	Age group targeted	25 - 34	Compare target vs actual engagement
target_interests	Topics/interests targeted	travel, fashion	Check match with actual user interests

Table 3: campaigns

Purpose: Contains campaign-level information (budget, duration, strategy).

Usage: Enables budget tracking, pacing, and ROI analysis.

Field	Description	Example	Use in Analysis
campaign_id	Unique campaign ID	11	Primary key; joins to ads
name	Campaign name	Campaign_41_Winter	Reporting, filtering
start_date	Campaign launch date	2025-03-01	Track active campaigns
end_date	Campaign end date	2025-03-31	Campaign duration analysis
duration_days	Derived: campaign length	30	Compare pacing & performance
total_budget	Budget allocated for campaign	39953.19	Basis for CPM, CPC, ROAS

Table 4: users

Purpose: Demographic and interest details of users engaging with ads.

Usage: Helps measure audience targeting accuracy (e.g., Ads targeted at women 18–24 vs actual engagement from men 25–34).

Field	Description	Example	Use in Analysis
user_id	Unique user identifier	141e5	Primary key; joins to ad_events
user_gender	Gender of user	Male	Gender-based performance
user_age	Age of user	27	Basis for custom segmentation
age_group	Grouped age bucket	25–34	Compare audience engagement by age
country	User’s country	India	Country-level reach analysis
location	More specific location (city/state)	Lisaport	Geo-targeting
interests	User’s interests	fashion, news	Match vs targeting interests

4. How the Tables Work Together

- **ad_events** → **ads**: Links events to ad details (platform, type, targeting).
- **ads** → **campaigns**: Links ads to campaign metadata (budget, duration).
- **ad_events** → **users**: Links user engagement events to demographic data.

This creates a star schema:

- **Fact Table**: ad_events
- **Dimension Tables**: ads, campaigns, users